Batch Simulation Report

GEMS Team

August 31, 2025 - 08:39

Abstract

This document contains a technical and epidemiological report of a batch run carried out using the **G**erman **E**pidemic **M**icro-Simulation **S**ystem (GEMS) v0.4.3. Code available at: https://gitlab.rlp.net/optim-agent/gems

Contents

Batch Information **1**

Simulation Results **1**
- Overview . . . 2
- Cases per Tick . . . 2
- Effective Reproduction Number . . . 2
- Total Tests per Tick . . . 3

Debug Information **3**
- Repository . . . 3
- System Information . . . 3
- Processor . . . 3
- Memory . . . 4

Glossary **4**

Batch Information

Table 1: Batch Information

Property	Value
Batch ID	2ca5b941-b251-4c7b-b4bd-044af179d274
Number of runs	100

Simulation Results

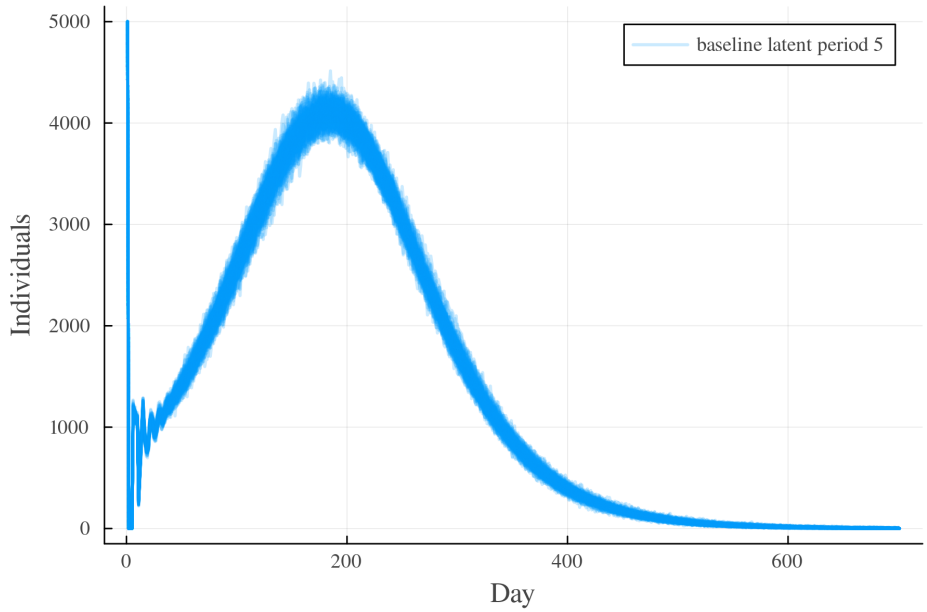
This section reports on data that was generated during the simulation execution.

Overview

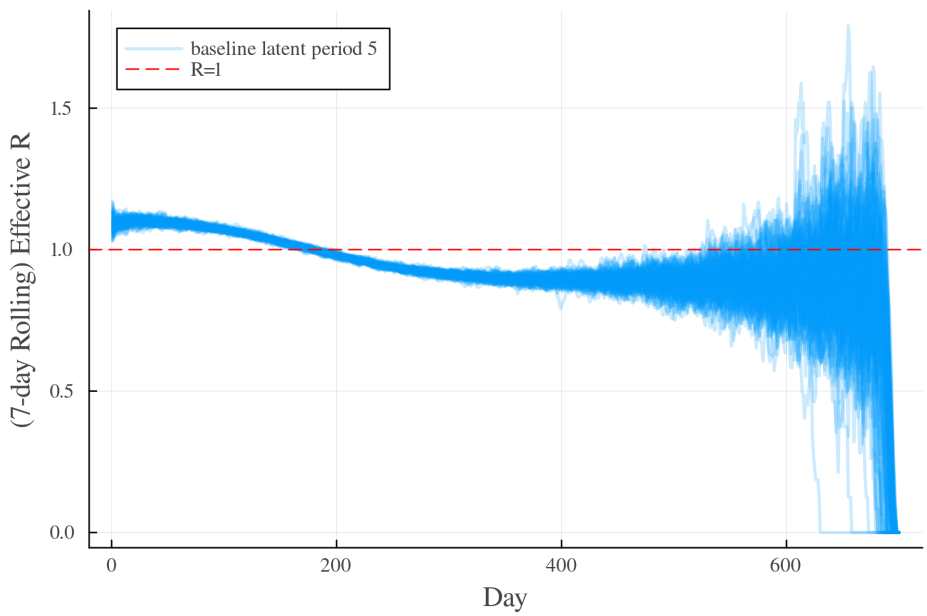
Table 2: Simulation Results Overview

Property	Value
Total Infections	929785.0, std: 7734.9, 95-CI: [928250.2-931319.8], range: [907484.0-947143.0]
Attack Rate[1]	18.6%, std: 0.15%, 95-CI: [18.57-18.63%], range: [18.15-18.94%]

Cases per Tick

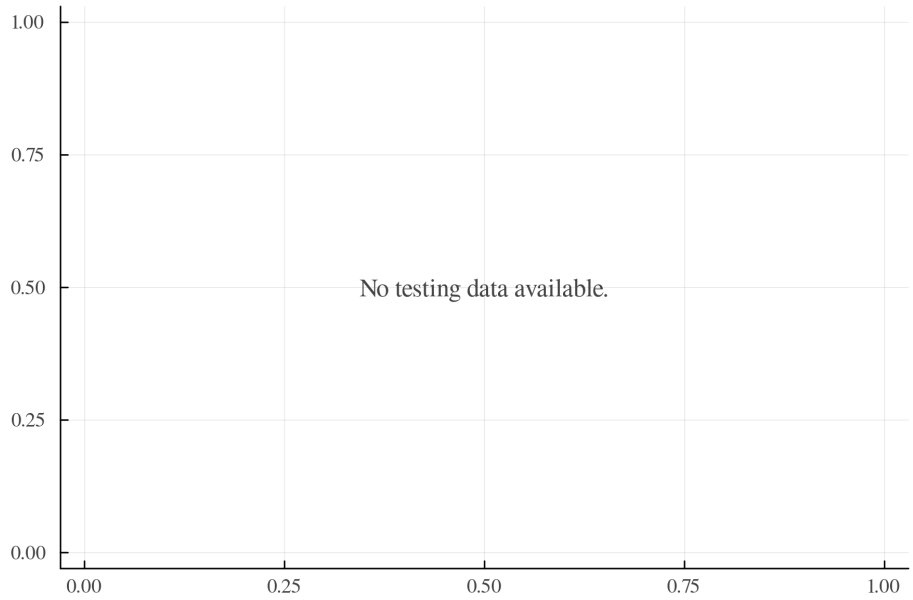


Effective Reproduction Number



[1]Total infections divided by number of individuals

Total Tests per Tick



Debug Information

Repository

Table 3: Repository Properties

Repository Property	Value
Origin Repository	No repository information available.
Branch	No branch information available.
Commit ID	No commit information available.

System Information

Table 4: General Sytem Properties

System Property	Value
System Kernel	Linuxx86_64-linux-gnu
Julia Version	1.11.5
Word Size	64
Julia Threads	32

Processor

Cpu Property	Value
Brand	Intel(R) Xeon(R) CPU E7- 4830 @ 2.13GHz
Vendor	:Intel
Architecture	:Westmere

Cpu Property	Value
Model	Family: 0x06, Model: 0x2f, Stepping: 0x02, Type: 0x00
Cores	8 physical cores, 16 logical cores (on executing CPU) Hyperthreading hardware capability detected
Clock Frequencies	Not supported by CPU
Data Cache	Level 1:3 : (32, 256, 24576) kbytes 64 byte cache line size
Address Size	48 bits virtual, 44 bits physical
SIMD	128 bit = 16 byte max. SIMD vector size
Time Stamp Counter	TSC is accessible via `rdtsc` TSC runs at constant rate (invariant from clock frequency)
Perf. Monitoring	Performance Monitoring Counters (PMC) revision 3 Available hardware counters per logical core: 3 fixed-function counters of 48 bit width 4 general-purpose counters of 48 bit width
Hypervisor	No

Memory

Memory Property	Value
Total Available Memory	451,447.47 MB
Free Memory	438,283.16 MB

Glossary

Term	Definition
Asymptomatic	A symptom category. An individual will not develop any symptoms. Its terminal state is "Presymptomatic".
Critical (State)	A state in the natural disease progression. An individual is in a critical condition with the need of hospitalization and possible need of ventilation or ICU.
Critical (Symptom Category)	A symptom category. The individual will get into a critical condition during its disease progression with the need of hospitalization and possible need of ventilation.
Critical Death Rate	The probability of individuals with critical condition to die at the end of the natural disease progression.
Disease Progression	An individual's course through the disease states from initial exposure to recovery or death.
Disease Status	The status of an individual in the natural disease progression. Holds information about infectiousness and symptoms.
Exposed	The individual is infected, but is not yet infectious.
Exposure	Exposure to a pathogen is regarded as an infection event and marks the inception of a disease progression.
Generation Time	The time interval between the infections of the infector and infectee, two immediate successors in an infection chain.

Term	Definition
Hospitalization Status	States if an individual is hospitalized and if additional measures have to be applied (vetilation, ICU).
Hospitalization Rate	The probability of an individual with severe symptoms to need hospitalization.
ICU	Short for Intensive Care Unit. An hospitalization state. Individuals are in the hospital, ventilated and are in ICU.
ICU Rate	The probabilityof an individual in critical condition to need ICU.
Individual	The representation of a person. It has different attributes and can become infected inside settings.
Infection Rate	Basic risk of getting infected upon infectious contact
Infectious Offset	An offset from the onset of symptoms. Determines if an individual becomes infectious before becoming symptomatic. Is also used for asymptomatic cases to determine the start of infectiousness.
Infectious	The individual is infectious, if it can infect other individuals.
Infectiousness	Measurement of how infectious an individual is. Reminiscent of viral load.
(Intervention) Measure	A single intervention-related activity chaning one distinct state variable in the simulation model (e.g. closing a particular school).
(Intervention) Strategy	A collection of intervention measures which are executed in chronological order.
(Intervention) Trigger	An event that causes an intervention strategy to be executed (e.g. an individual starting to experience symptoms).
Isolation	As of now defined as “household isolation”. An individual staying in its household while still being able to have contacts to other household members.
Length of Stay	The time an individual will spend inside the hospital.
Mild	An individual will only develop mild symptoms. Its terminal state is “Symptomatic”.
Mild Death Rate	The probability of individuals with mild symptoms to die at the end of the natural disease progression.
Onset of Symptoms	The time when symptoms start. Is generally calculated and given as an increment to the tick, when an infection happend.
Onset of Severeness	The time when symptoms become severe as an increment to the onset of symptoms.
Pathogen	An enclosure to collect disease-related parameters such as infection rates or the delay until the onset of symptoms.
Presymptomatic	A state in the natural disease progression. The individual is infected but does not experience symptoms yet.
Removed	The state of an individual being recovered or dead
Setting	Some sort of context in which contacts can happen. Those can be Households or Offices or more abstract settings like social networks.
Severe (State)	A state in the natural disease progression. The individual has severe symptoms with the potential need to be hospitalized.

Term	Definition
Severe (Symptom Category)	A symptom category. An individual will develop severe symptoms and has possibly the need to be hospitalized. Its terminal state is “Severe”.
Severe Death Rate	The probability of individuals with severe symptoms to die at the end of the natural disease progression.
Symptom Category	Categorization of the disease progression regarding the occuring symptoms.
Symptomatic	A state in the natural disease progression. The individual has symptoms, but they aren’t severe.
Terminal State	The last state reached by an individual in the disease progression before being removed
TestType	Parameterizing a certain type of test (e.g. PCR or Antigen).
Test Sensitivity	A test’s ability to positively identify an infected individual.
Time to Hospitalization	If an individual will need hospitalization eventually, this defines the time between the onset of symptoms and hospitalization.
Time to ICU	If an individual will need ICU eventually, this defines the time between the time to hospitalization and the delivery into ICU.
Time to Recovery	The time it takes an individual to recover from an infection as an increment from the onset of symptoms. If an individual will die at the end of the natural disease progression, this will (also) define the time of death.
Ventilated	An hospitalization state. Individuals are in the hospital and receive ventilation.
Ventilation Rate	The probability of an individual in critical condition to need ventilation.